Assignment 5

Total = 10 Marks

Multiple Choice Questions

Question 1: A two-dimensional copula is defined on:

(a) $(-\infty, \infty)$
(b) $[0, \infty)$
(c) $[0,1]^2$
(d) $[0,1]$

[1 *Mark*]

Correct answer: (c) $[0, 1]^2$

Detailed Solution:

A copula is a function that links the marginal cumulative distribution functions (CDFs) of random variables to form their joint distribution. Since each marginal CDF takes values between **0 and 1**, the inputs to a copula are the transformed variables $u = F_X(x)$ and $v = F_Y(y)$, where $u, v \in [0,1]$.

Therefore, a two-dimensional copula is defined on the unit square

$$[0,1] \times [0,1] = [0,1]^2.$$

Hence, the correct answer is: $[0, 1]^2$.

Question 2: The H-volume of a valid 2-dimenssional copula for every rectangle within $[0,1] \times [0,1]$ domain must be

(a) Negative
(b) Positive
(c) Non-negative
(d) Non-positive

[1 *Mark*]

Correct answer: (c) Non-negative

Detailed Solution:

The H-volume of a rectangle represents the probability assigned by a copula to that rectangle in the unit square $[0,1]^2$. Since probabilities can never be negative, the H-volume of every rectangle must always be non-negative.

Therefore the correct answer is Non-negative.

Question 3: The upper Frechet-Hoeffding bound is

(a) uv
(b) $\min(u, v)$
(c) $\max(u + v - 1, 0)$

(d) $1 - u - v$

[1 *Mark*]

Correct answer: (b) min (u, v)

Detailed Solution:

The Fréchet–Hoeffding bounds define the minimum and maximum possible values that any bivariate copula $C(u, v)$ can take. Every valid copula must satisfy

$$\boldsymbol{max\ (u + v - 1, 0) \leq\ C(u, v)\ \leq\ min\ (u, v)}$$

The functions $\boldsymbol{max\ (u + v - 1)}$ and $\boldsymbol{min(u, v)}$ are called lower and upper Frechet-Hoeffding bounds respectively.

Therefore the correct answer is $\boldsymbol{min\ (u, v)}$.

Question 4: Which of the following functions is a valid copula?

(a) $C(u, v) = u^2 + v^2$
(b) $C(u, v) = |u + v - 2|$
(c) $C(u, v) = \frac{1}{u} + \frac{1}{v}$
(d) $C(u, v) = uv$

[1 *Mark*]

Correct answer: (d) $\boldsymbol{C(u, v) = uv}$

Detailed Solution:

For $C(u, v) = u^2 + v^2$, $C(u, 1) = u^2 + 1 \neq 1$, It is not a valid copula.
For $C(u, v) = |u + v - 2|$, $C(u, 0) = |u - 2|$, It is not grounded, so it's not a valid copula.
For $C(u, v) = \frac{1}{u} + \frac{1}{v}$, $C(u, 0) = \frac{1}{u} + \frac{1}{0} \neq 0$. It is not grounded. Hence, it is not a valid copula.
For $C(u, v) = uv$, $C(u, 0) = 0, C(0, v) = 0$, and $C(u, 1) = u, C(1, v) = v$. So, $C(u, v) = uv$ satisfied grounded and 2-increasing both conditions. This is valid copula.

Therefore the correct answer is $\boldsymbol{C(u, v) = uv}$.

Question 5: The following paired observations of two random variables X and Y are given.

Observation	**1**	**2**	**3**	**4**	**5**
$\boldsymbol{X}$	12	16	20	25	30
$\boldsymbol{Y}$	18	15	24	21	27

Calculate Kendall's Rank Correlation Coefficient (Kendall's Tau)**.**

(a) 0.5
(b) 0.6
(c) 0.7
(d) 0.8

[3 *Mark*]

Correct answer: (b) 0.6

Detailed Solution:

Observation	**1**	**2**	**3**	**4**	**5**
$\boldsymbol{X}$	12	16	20	25	30
$\boldsymbol{Y}$	18	15	24	21	27

The corresponding ranks are given by

$\boldsymbol{R^X}$	1	2	3	4	5

R^Y	2	1	4	3	5

Rank Pair (R^X, R^Y)	**Concordant** (C)	**Discordant** (d)
(1,2)	3	1
(2,1)	3	0
(3,4)	1	1
(4,3)	1	0
(5,5)	0	0
Total	8	2

Number of pairs $(n_{C_2}) = 5_{C_2} = 10$

$$\text{Kendall's Tau} = \tau = \frac{c-d}{n_{C_2}}$$

$$\tau = \frac{8-2}{10} = 0.6$$

Therefore, the correct answer is 0.6.

Question 6: For $u = 0.25$ and $v = 0.80$, a candidate copula gives $C(u, v) = 0.95$. According to the Fréchet–Hoeffding bounds, this value is

(a) Valid
(b) Invalid
(c) Depends on the marginals
(d) Cannot be determined

[3 *Mark*]

Correct answer: (b) Invalid

Detailed Solution:

The Fréchet–Hoeffding bounds specify the permissible range of values that any copula $C(u, v)$ can take. For every $(u, v) \in [0,1]^2$,

$$max(u + v - 1,0) \leq C(u, v) \leq \min(u, v)$$

Given $u = 0.25$ and $v = 0.80$, the candidate copula value is $C(u, v) = 0.95$.

For lower bound $max(u + v - 1,0) = max(0.25 + 0.8 - 1,0) = 0.05$.

For upper bound $min(u, v) = min(0.25, 0.80) = 0.25$.

The candidate value $(C(u, v) = 0.95)$ does not lie between the Fréchet–Hoeffding bound, so it is Invalid.

Therefore, The correct answer is Invalid .